

花瓣仔與均衡飲食城

第1頁 PAGE 1

PETALINO AND THE CITY OF BALANCED DIET

能量平衡機 / THE ENERGY BALANCE MACHINE

1

我來到
均衡飲食城！

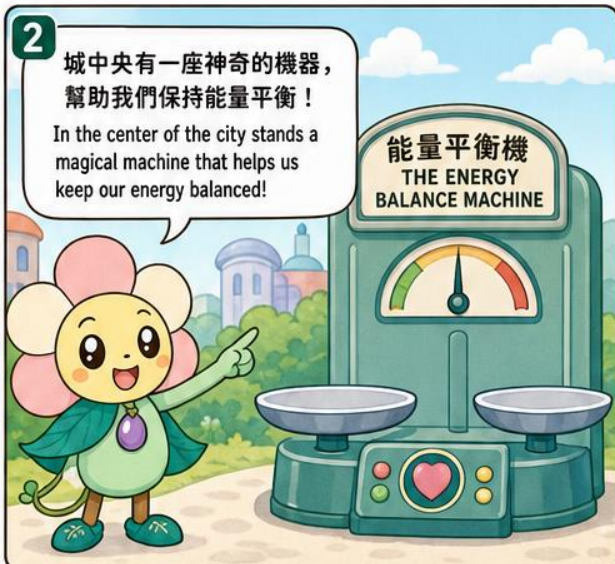
I've arrived at the
City of Balanced
Diet!



2

城中央有一座神奇的機器，
幫助我們保持能量平衡！

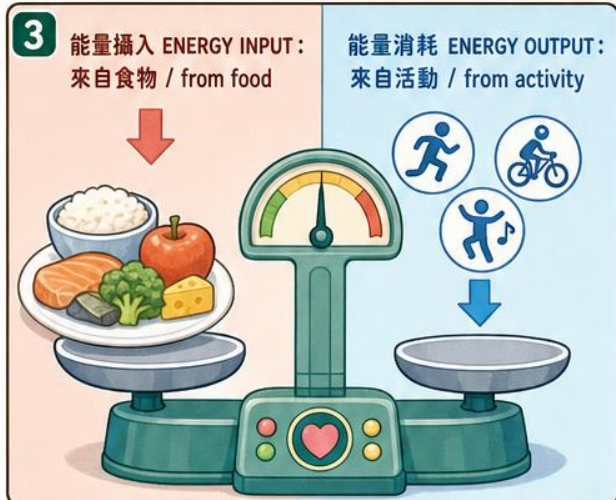
In the center of the city stands a
magical machine that helps us
keep our energy balanced!



3

能量攝入 ENERGY INPUT:
來自食物 / from food

能量消耗 ENERGY OUTPUT:
來自活動 / from activity



4

均衡飲食：能量攝入與能量消耗之間取得平衡

Balanced diet: balance between
energy input and energy output.



5

均衡飲食不只是吃不同食物，
也要平衡能量攝入和消耗！

A balanced diet is not only about
food variety—it must balance
energy input and output!



6

準備好和我一起探索更多健康秘密了嗎？

下一站，我們出發！

Ready to explore more health secrets with me?
Next stop, let's go!



營養不良的兩面

THE TWO SIDES OF MALNUTRITION

1

營養不良有兩種方向，讓我們一起來了解！
Malnutrition has two directions. Let's explore them together!



營養不良
MALNUTRITION

營養不足
UNDERNUTRITION

營養過剩
OVERNUTRITION

2

能量攝入 > 能量消耗
ENERGY INPUT > ENERGY OUTPUT



能量攝入
ENERGY INPUT

能量消耗
ENERGY OUTPUT

3

多餘能量儲存在脂肪組織，生物量增加，體重隨時間上升
Excess energy is stored as fat in adipose tissue; biomass increases and weight rises over time.



脂肪組織（能量儲存）
ADIPOSE TISSUE (ENERGY STORAGE)

4

能量消耗 > 能量攝入
ENERGY OUTPUT > ENERGY INPUT



能量攝入
ENERGY INPUT

能量消耗
ENERGY OUTPUT

5

身體分解儲存的糖原和脂肪以釋放能量，生物量減少，體重隨時間下降
The body breaks down stored glycogen and fat to release energy; biomass decreases and weight falls over time.



糖原儲存
GLYCOGEN
STORAGE

脂肪儲存
FAT
STORAGE



6

重點回顧
RECAP



攝入 > 消耗
Input > output



儲存脂肪
Fat storage



體重增加
Weight gain



消耗 > 攝入
Output > input



分解儲備
Stored food
broken down



體重減輕
Weight loss



保持平衡，
助你健康成長！
Balance is key
for healthy
growth!



均衡飲食與
活躍生活，
讓身體運作
更棒！
Eat well and stay
active for a
stronger you!

第3頁 PAGE 3 — 成長中的身體需要更多 GROWING BODIES NEED MORE

1

歡迎來到成長區！
一起看看誰的身體
需要更多營養吧！

Welcome to
Growth District!
Let's see whose
bodies need
more nutrients!



2

兒童需要較多能量：
身體較小，散失熱量較多
Children need more energy:
greater heat loss.



3

成長需要特別的營養素幫助！
Growth needs special nutrients!

蛋白質
PROTEIN



快速生長，
例如肌肉
rapid growth,
e.g. muscle

鈣 + 維生素D
CALCIUM +
VITAMIN D



生長骨骼和牙齒
grow bones
and teeth

鐵
IRON



製造血紅蛋白
make haemoglobin
in blood

4

這些額外的營養需求，
適用於兒童和胎兒（孕婦）！
These increased needs apply to
CHILDREN / FOETUS (PREGNANT WOMEN.)!



兒童
CHILDREN

胎兒（孕婦）
FOETUS (PREGNANT WOMEN)

5

孕婦需要更多膳食纖維，
預防便秘。

Pregnant women need more
dietary fibre to prevent
constipation.



6

成長越快，
建造身體需要的
材料越多！
Rapid growth needs
more building
materials!



均衡飲食，滿足成長所需，讓身體茁壯成長！
Eat a balanced diet to meet growth needs and help our bodies thrive!

第4頁 PAGE 4 — 長者的營養需要 CARING FOR THE ELDERLY

1

營養對長者的健康和活力非常重要！

Nutrition is very important for the health and vitality of our elderly!

謝謝你來探望我，和我分享營養知識！

Thank you for visiting me and sharing nutrition knowledge!

2

較多鈣和維生素D：預防骨質疏鬆

More calcium and vitamin D: prevent osteoporosis.



強壯骨骼
Strong bones



脆弱骨骼
Fragile bones



適量曬太陽，多攝取含鈣和維生素D的食物，有助維持骨骼健康！

Moderate sunlight and foods rich in calcium and vitamin D help maintain healthy bones!

Ca

鈣

CALCIUM

+

D

維生素D

VITAMIN D

3

較少能量：活動量通常較低

Less energy: generally less active.



選擇適合年齡和活動量的均衡飲食，提供身體所需營養，不必過量！

Choose a balanced diet suitable for age and activity level—provides what the body needs, without overeating!

4

均衡飲食要配合年齡和活動量！

A balanced diet must match age and activity level!



5

重點回顧 RECAP



鈣 + 維生素D

CALCIUM + VITAMIN D

預防骨質疏鬆

Prevent osteoporosis.



較少能量

LESS ENERGY

活動量通常較低

Generally less active.

讓長者每天吃得好、動得好、生活更美好！

Help the elderly eat well, stay active, and live well every day!



關心長者，從營養開始！

Care starts with good nutrition!



1 較少能量：久坐生活方式
Less energy: sedentary lifestyle



更多膳食纖維和水：預防便秘
More dietary fibre and water: prevent constipation.



膳食纖維
Fibre



水分
Water

多菜、多全穀、
多豆類，加上足夠水分！
Eat more vegetables, whole grains,
legumes, and drink enough water!



2 更多能量：支援肌肉活動
More energy: supports muscular movement



能量
Energy



肌肉
Muscle

足夠碳水化合物和優質蛋白質，
支援肌肉活動與恢復！
Enough carbohydrates and quality protein
to support muscular activity and recovery!



3 更多維生素D：日照不足
More vitamin D: not enough sunlight



維生素D
Vitamin D



食物來源
Food sources

選擇維生素D食物，
讓陽光有機會照到你！
Choose vitamin D-containing foods,
and get safe sunlight when you can!



相同食物
不同份量；
需要取決於
生活方式！
Similar foods,
different amounts—
needs depend on
lifestyle!



久坐生活方式
Sedentary lifestyle



支援肌肉活動
Muscular activity



日照不足
Low sunlight



了解自己的生活方式，作出聰明的食物選擇！
Know your lifestyle, make smart food choices!

第6頁 PAGE 6 — 男女的營養需要

NUTRITIONAL NEEDS OF MALES AND FEMALES

1

今天我們來比較
兩個一般的例子！
Today, let's compare
two general cases!



2

男性的一般情況 / MALE CASE

男性通常需要較多能量
和蛋白質：一般肌肉量較多
Males generally need more
energy and protein:
usually more muscular.



3

能量提供日常活動所需，
蛋白質有助身體生長和修復。

Energy fuels daily activities,
and protein helps the body
grow and repair.



提供能量
FUEL FOR
ACTIVITIES



幫助生長與修復
HELPS GROWTH
AND REPAIR

4

女性青春期的一般情況 / FEMALE CASE (PUBERTY)

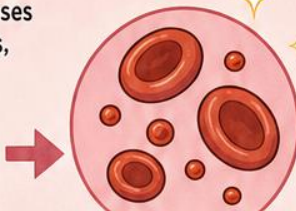
青春期女性需要較多鐵

Females during puberty
need more iron.



5

月經造成定期失血，
Menstruation causes
regular blood loss,



紅血球 (含血紅素)
RED BLOOD CELLS
(HAEMOGLOBIN)

因此需要較多鐵來製造血紅素，
以維持足夠的氧氣輸送。
so more iron is needed to make
haemoglobin and keep oxygen
transport adequate.

6

營養需要受年齡、性別、活動量
和生活方式影響

Nutritional needs vary with age, sex,
activity and lifestyle.



需要因人而異！
Needs differ
between
individuals!



以上為一般情況，每個人的需要可能不同，請諮詢合資格的專業人士。

The above are general patterns. Individual needs may differ—please consult a qualified professional.

第7頁 PAGE 7 — 重建食物金字塔 (下層)

REBUILDING THE FOOD PYRAMID: LOWER LEVELS

1

從下層開始，
一層一層把金字塔
建得更穩固！

Let's build from the
bottom up and make
the pyramid strong!



2

穀物 GRAINS

每天6-8份 / 6-8 servings per day

所佔最多，主要能量來源 /
largest proportion, main source of energy.



3

蔬菜 VEGETABLES

每天最少3-5份 /
at least 3-5 servings per day



含豐富維生素和礦物質 / rich in vitamins and minerals

4

水果 FRUITS

每天2-4份 / 2-4 servings per day

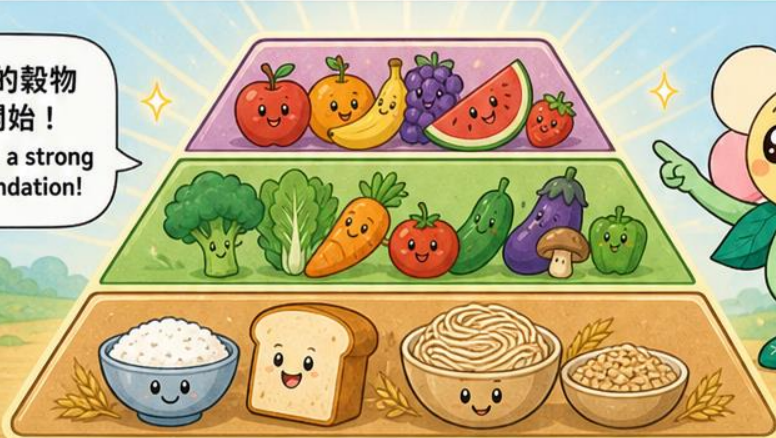


含豐富維生素和礦物質 / rich in vitamins and minerals

5

由穩固的穀物
基礎開始！

Start with a strong
grain foundation!



穩固的下層，支撐健康的每一天！ / A strong base supports a healthy every day!

6 糖和鹽 SUGARS AND SALT

進食量最少
smallest amounts

增添味道但營養價值低；
過量會損害健康
provide flavour but little
nutritional value;
excess can harm health.



5 蛋白質食物 PROTEIN FOODS

(肉、魚、蛋、豆類 /
meat, fish, eggs, beans) :

蛋白質供生長和修補
proteins for growth
and repair.



4 奶類 DAIRY

每天 2-3 份 /
2-3 servings per day —
提供鈣和維生素 D /
provides calcium
and vitamin D.



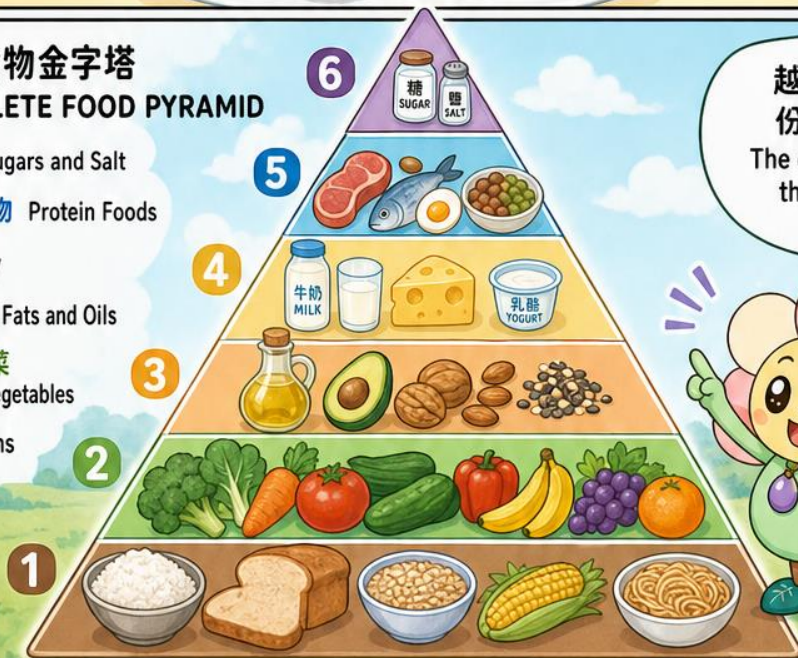
3 脂肪和油 FATS AND OILS

提供必需脂肪酸 /
provide essential
fatty acids.



7 完整食物金字塔 THE COMPLETE FOOD PYRAMID

- 6 糖和鹽 Sugars and Salt
- 5 蛋白質食物 Protein Foods
- 4 奶類 Dairy
- 3 脂肪和油 Fats and Oils
- 2 水果和蔬菜 Fruits and Vegetables
- 1 穀類 Grains



越接近頂部，
份量越少！

The closer to the top,
the smaller the
amount!



第9頁 PAGE 9 — 食物標籤計算挑戰

THE FOOD-LABEL CHALLENGE

PETALINO
BIOLOGINO

快看！包裝上有
營養標籤資料！
Look! There's nutrition
information on the package!



任務卡 MISSION CARD

需要取得
10克目標營養素

Need to obtain
10 g of the target nutrient.



1 把每100克食物視為一份 Treat every 100 g as one portion.



2 計算所需份數 Calculate portions needed

$$10 \text{ g} \div 1.3 \text{ g} = 7.69 \text{ 份} / \text{portions}$$



← 所需份數（可為小數）
Portions needed
(can be a decimal)

3 換算成實際食物重量 Convert to actual food mass

$$7.69 \times 100 \text{ g} = 769.23 \text{ g}$$

答案修正至
小數點後兩位
Give the answer
correct to 2
decimal places.



769.23 g 是本次任務在此條件下的「計算結果」，
並非建議每日食用份量。

769.23 g is the calculated amount for this example,
not a recommended daily serving.



本次計算結果
Calculated amount
(769.23 g)

≠



建議每日食用份量
Recommended daily
serving

第10頁 PAGE 10 — BMI陷阱與任務總結

THE BMI TRAP AND FINAL MISSION

1 BMI陷阱 THE BMI TRAP

BMI沒有考慮身體組成，例如脂肪和肌肉百分比。
BMI does not account for body composition, such as fat and muscle percentages.



肌肉較多 More Muscle

BMI 22.0

肌肉 %
Muscle %
50%



脂肪 %
Fat %
15%



BMI 22.0



脂肪較多 More Fat

肌肉 %
Muscle %
25%



脂肪 %
Fat %
35%



同樣的BMI，
身體組成
可能大不同！
Same BMI,
different body
composition!



2 最終任務：恢復能量平衡機與食物金字塔！ FINAL MISSION: RESTORE THE ENERGY BALANCE MACHINE AND THE FOOD PYRAMID!



食物金字塔
FOOD PYRAMID



3 任務總結 MISSION RECAP

1 平衡能量
攝入與消耗
Balance energy
input and output



2 營養需要
因人而異
Nutritional needs
differ among
individuals



3 依照食物金字塔
選擇合適份量
Follow appropriate
food-pyramid portions



4 糖、鹽和脂肪
避免過量；
BMI不能單獨
判斷健康
Avoid excess sugar,
salt and fat; do not use
BMI alone to judge
health.



4 沒有一種餐單適合所有人——
真正的均衡飲食要配合個人需要！
One diet does not fit everyone—
a truly balanced diet matches
individual needs!



均衡每一天，健康每一步！
Balanced every day,
healthy every step!



任務完成！
MISSION COMPLETE!